

NanoSafe Analyzer — Research & Cytotoxicity Report

In Vitro Evaluation of ZnO Nanoparticle Biocompatibility | ISO 10993-5 Compliance Assessment

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1. EXPERIMENT OVERVIEW

Experiment / Sample Name: PDF Test 1
Cell Line Evaluated: HeLa
Medical Application: General Biomedical
Lead Researcher: Researcher

Nanoparticle Type: ZnO
Exposure Duration: N/A
ISO 10993-5 Compliance: FAIL — Cytotoxic
Data Source: Direct Entry

2. STUDY PARTICIPANT & BIOLOGICAL SAMPLE TRACEABILITY

Biological Sample: Standard in-vitro reference cell culture line. Anonymized laboratory experiment.

3. AVERAGED CYTOTOXICITY & BIOMARKER METRICS

Mean Cell Viability: 71.7%
Reactive Oxygen Species (ROS): 2.84x
Apoptosis Inducement: 6.14%

Mean ZnO Concentration: 37.0 µg/mL
LDH Membrane Leakage: 9.459999999999999%

4. MACHINE LEARNING CYTOTOXICITY ASSESSMENT

Predicted Toxicity Score: 11.51 / 100
Estimated IC50 (4PL Curve): 58.01 µg/mL
ML Model Confidence: 98.5%

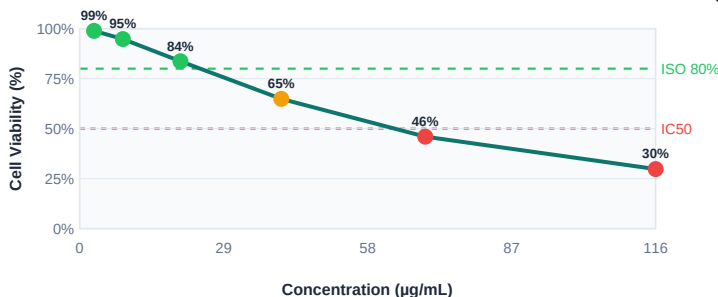
Risk Classification: Low
Biocompatible Safe Window: 0.0 - 15.7 µg/mL

5. SCIENTIFIC INTERPRETATION & RECOMMENDATION

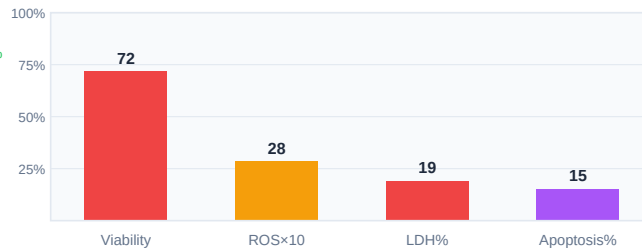
Local Scikit-Learn Ensemble Model evaluated ZnO cytotoxicity on HeLa cells under a 24.0h exposure duration for 'Wound Dressing & Topical Antimicrobial'. Measured parameters: Viability=71.7%, ROS=2.84, LDH=9.459999999999999%, Apoptosis=6.14%. The offline ML model computed a Toxicity Score of 11.51 out of 100 (Low Risk Zone, ISO 10993-5 status: CONDITIONAL — Low-Dose Monitoring Required). Predicted IC50: 44.83

6. DOSE-RESPONSE CURVE & BIOMARKER PROFILE CHARTS

Graph A: Cell Viability vs. Concentration (Dose-Response Curve)



Graph B: Multi-Biomarker Safety Profile



7. CLINICAL SUGGESTIONS & NEXT STEPS

1. CAUTION — MODERATE TOXICITY:

Cell viability at 71.7% is below ISO 10993-5 PASS threshold. Formulation requires optimization before biomedical use.

2. DOSE REDUCTION:

Reduce working concentration by 40-60% from current maximum. Target viability $\geq 80\%$ at all therapeutic dose points. Current safe ceiling: 0.0 - 15.7 µg/mL.

3. IC50 MARGIN:

With IC50 at 58.0 µg/mL, reduce maximum application dose to 17.4 µg/mL (30% of IC50) to maintain adequate safety margin.

4. SURFACE MODIFICATION:

Evaluate citrate or PVP surface functionalization to reduce ROS generation and membrane oxidative stress. Repeat cytotoxicity assay after coating.

5. REPEAT STUDY:

Conduct 3-replicate repeat assay with optimized formulation before proceeding. Include positive control (0.1% Triton X-100) and negative control (PBS) in each plate.